

CS50: Artificial Intelligence with Python

Detailed Self■Study Notes with Explanations, Diagrams & Examples

Introduction to Artificial Intelligence

Artificial Intelligence (AI) is the field of computer science focused on creating systems that can perform tasks requiring human■like intelligence such as reasoning, learning, problem■solving, perception, and language understanding.

Goals of AI:

- Build intelligent agents
- Automate decision■making
- Mimic human reasoning and learning
- Solve complex real■world problems

1. Search Algorithms

Search problems involve finding a sequence of actions that leads from an initial state to a goal state. AI agents explore state spaces using search algorithms.

Key Components:

- State – representation of the world
- Initial State
- Actions
- Transition Model
- Goal Test
- Path Cost

Algorithms Explained:

- Depth-First Search (DFS): explores deep paths first
- Breadth-First Search (BFS): explores level by level
- Uniform Cost Search (UCS): expands lowest cost first
- Greedy Best-First Search: uses heuristic only
- A* Search: uses cost + heuristic (optimal & complete)

Diagram Concept: Tree structure showing nodes, branches, frontier, explored set.

2. Knowledge Representation & Logic

AI systems use logic to represent knowledge and draw conclusions. Propositional logic uses true/false statements to represent facts.

Key Ideas:

- Propositional symbols
- Logical connectives (AND, OR, NOT, IMPLIES)
- Knowledge Base
- Inference
- Model Checking

3. Uncertainty & Probability

Real-world environments are uncertain. AI uses probability to reason under uncertainty.

Concepts:

- Random Variables
- Conditional Probability
- Bayes' Theorem
- Bayesian Networks

Bayesian Networks are graphical models representing probabilistic relationships using directed acyclic graphs (DAGs).

4. Machine Learning

Machine Learning enables systems to learn patterns from data rather than being explicitly programmed.

Types of Learning:

- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning

Algorithms:

- k-Nearest Neighbors
- Perceptron
- Support Vector Machines

5. Neural Networks

Neural networks are inspired by the human brain and consist of layers of interconnected neurons.

Structure:

- Input Layer
- Hidden Layers
- Output Layer

Training involves forward propagation, loss calculation, and backpropagation to adjust weights.

6. Natural Language Processing (NLP)

NLP enables machines to understand and generate human language.

Topics:

- Tokenization
- Markov Models
- Text Classification
- Language Models

Conclusion & Study Tips

This PDF is designed as long-term revision material. Re-read diagrams, revise definitions, and practice Python implementations for mastery.